

**B. Tech Degree V Semester (Supplementary) Examination
July 2009**

ME 505 POWER PLANT ENGINEERING

(2006 Scheme)

Time : 3 Hours

Maximum Marks : 100

PART – A

(Answer ALL questions)

(8 x 5 = 40)

- I. (a) What are the two basic parameters to decide while planning a power plant?
(b) What do you understand by depreciation? How is the depreciation rate determined?
(c) State the essential elements of a hydroelectric power plant.
(d) Give the specific advantages and disadvantages of a gas turbine plant for a utility system.
(e) Explain Nuclear Fission.
(f) Write short note on environmental aspects of a thermal power station.
(g) What are the advantages and limitations of tidal power generation?
(h) What is OTEC?

PART – B

(4 x 15 = 60)

- II. A power plant of 210 MW installed capacity has the following particulars.

Capital cost	=	Rs.18,000/- KW installed
Interest and depreciation	=	12%
Annual load factor	=	60%
Annual capacity factor	=	54%
Annual Running charges	=	Rs. 200 x 10 ⁶
Energy consumed by power plant axillaries	=	6%

Calculate

- (a) the cost of power generation per KWH
(b) the reserve capacity

(15)

OR

- III. A new factory requires a maximum demand of 700 KW and load factor of 25%.
The following two suppliers are available.

- (i) public supply tariff is Rs.48 per KW of maximum demand plus 2.4P. per KWH.

Capital cost	=	Rs.84000
Interest & depreciation	=	10%

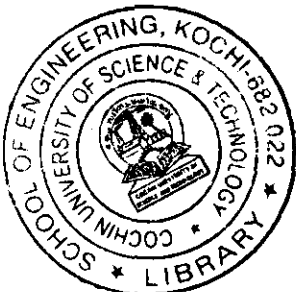
- (ii) private oil engine generating station

Capital cost	=	300000
Fuel consumption	=	3N/KWH
Cost of fuel	=	Rs.8.4 per KN
Wages	=	0.48P/KWH
Maintenance Cost	=	0.36P/KWH
Interest and depreciation	=	15%

Find which supply will be more economical.

(15)

(Turn Over)



IV. The following data refers to a proposed hydroelectric power plant.

Available head	=	27 m
Catchment area	=	430 Sq. Km.
Rainfall	=	150 cm/year
Percentage of total rainfall utilised	=	65%
Penstock efficiency	=	95%
Turbine efficiency	=	80%
Generator efficiency	=	86%
Load factor	=	0.45

- (i) Calculate the power developed.
- (ii) Suggest suitable turbines for the plant.
- (iii) Assume two machines are of equal capacity are provided, calculate the capacity of each unit. (15)

OR

V. A gas turbine power plant consists of a two stage compressor with intercooling and a single stage turbine with a regenerator. Air enters the compressor at 1 bar, 20°C . The maximum temperature of the cycle is limited to 900° and the maximum pressure ratio is 6. The effectiveness of the regenerator is 0.7. The rate of air flow through the plant is 210 Kg/s and the calorific value of fuel used is 40.8 MJ/Kg. The isentropic efficiency of both the compressors is 0.82, the isentropic efficiency of the turbine 0.92, the combustion efficiency is 0.95, the mechanical efficiency is 0.96 and the generator efficiency is 0.95. Take for air $C_p = 1.005 \text{ KJ/KgK}$ and $\gamma = 1.4$ and for gases $C_p = 1.08 \text{ KJ/KgK}$ and $\gamma = 1.33$. Assuming perfect intercooling and neglecting pressure and heat losses, estimate

- (i) the air-fuel ratio
- (ii) the cycle efficiency
- (iii) the power supplied by the plant
- (iv) the specific fuel consumption of the plant
- (v) the fuel consumption per hour. (15)

- VI. (a) How the steam power plants are classified? (5)
- (b) Explain the fuel handling system in a steam power plant. (10)

OR

VII. A reactor is fuelled with 100 tonnes of natural uranium (atomic mass 238.05) in which the average thermal neutron (2200 m/s) flux is 10^{13} neutrons/cm²s. The 2200 m/s cross section of U - 235 (atomic mass 235.04) are $\sigma_f = 579$ barns and $\sigma_c = 101$ barns. The energy release per fission is 200 MeV and 0.715% of natural uranium is U-235. Calculate :

- (i) the rating of the reactor in MW/tonne
- (ii) the rate of consumption of U - 235 per day. (15)

- VIII. (a) What are fuel cells? (6)
- (b) Explain with a neat diagram of a wind electric generating power plant. (9)

OR

- IX. (a) Write short notes on solar collectors. (6)
- (b) What is the total power of a wind stream? What do you mean by coefficient of performance and tip speed ratio of a windmill? (9)
